

Phu Gia Hoang

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SUMMARY

Mechanistic interpretability researcher developing causal and geometric methods to study how internal representations shape model behavior. Current work focuses on sparse autoencoders, feature interventions, and the reliability of model steering, with broader interests in technical alignment.

RESEARCH EXPERIENCE

Research Associate, Ubiquitous Knowledge Processing (UKP) Lab

Darmstadt, Germany

Academic advisors: Subhabrata Dutta and Iryna Gurevych

Feature-Effect Geometry Analysis (FEGA)

Jun 2025–Present

- Introduced FEGA, an unsupervised causal framework that ablates SAE features across contexts and analyzes the geometry of their downstream logit effects.
- Found that consistent 1D effects are rare: value-like features are structured and low-dimensional, while pointer-like features are diffuse and context-dependent, limiting the reliability of SAE features as steering directions.

M.Sc. Thesis Research, MBZUAI

Abu Dhabi, UAE

Academic advisors: Yova Kementchedjhieva and Kentaro Inui

Patch-Copying Repeated Tokens for Efficient Generation

Dec 2024–May 2025

- Designed and evaluated a training-free method for reusing internal representations across repeated token spans in Qwen2.5-3B; shallow layer reuse preserved outputs while skipping redundant computation.

The UIT NLP Group, UIT-VNUHCM

HCMC, Vietnam

Academic advisors: Kiet Nguyen Van and Ngan Luu-Thuy Nguyen

Hate Speech Spans Detection in Vietnamese

Mar 2021–Apr 2022

- Developed ViHOS, an 11,056-comment Vietnamese hate/offensive-span dataset with span-level annotations and benchmark experiments; published as first author at EACL 2023.

PUBLICATIONS

Phu Gia Hoang*, Anwoy Chatterjee*, Tanmoy Chakraborty, Iryna Gurevych, and Subhabrata Dutta. 2026. **Sparse Autoencoders Encode Both Concepts and Functions: The Downstream Geometry of Feature Effects.** *arXiv preprint; under review at JMLR*. [Paper] [Project] [Code]

Phong Do, Son Tran, Phu Hoang, Kiet Nguyen, and Ngan Nguyen. 2024. **VLUE: A New Benchmark and Multi-task Knowledge Transfer Learning for Vietnamese Natural Language Understanding.** *Findings of NAACL 2024*. [Paper] [Project]

Phu Gia Hoang, Canh Duc Luu, Khanh Quoc Tran, Kiet Van Nguyen, and Ngan Luu-Thuy Nguyen. 2023. **ViHOS: Hate Speech Spans Detection for Vietnamese.** *EACL 2023*. [Paper] [Code]

Phu Gia Hoang, Le Thanh, and Hai-Long Trieu. 2023. **VBD_NLP at SemEval-2023 Task 2: Named Entity Recognition Systems Enhanced by BabelNet and Wikipedia.** *SemEval 2023*. [Paper]

Khanh Quoc Tran, An Trong Do, Phu Gia Hoang, Canh Duc Luu, Trong Hop Do, and Kiet Van Nguyen. 2022. **Vietnamese Hate and Offensive Detection using PhoBERT-CNN and Social Media Streaming Data.** *Neural Computing and Applications*. [Paper] [Code]

Technical Writing

Phu Gia Hoang. 2026. **Can We Teach a Model to Encode a Semantic Feature on a Chosen Manifold in Just Three Channels?** *LessWrong*. [\[Post\]](#) [\[Puzzle Write-up\]](#) [\[Code\]](#)

EDUCATION

M.Sc. in Natural Language Processing

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI)

GPA: 3.57/4.0 (Full sponsorship)

Thesis: Patch-Copying Repeated Tokens for Efficient Generation

Abu Dhabi, UAE

Aug 2023–May 2025

B.Sc. in Data Science

University of Information Technology - Vietnam National University, HCMC

GPA: 8.89/10 (2nd in graduating class)

Thesis: Hate Speech Spans Detection in Vietnamese (10/10)

HCMC, Vietnam

Aug 2019–Apr 2023

HONORS & AWARDS

Honourable Mention, [BlueDot Technical AI Safety Puzzle #1](#), BlueDot Impact 2026

Consolation Prize, 23rd Scientific Research – Eureka, Ho Chi Minh Communist Youth Union & Vietnam National University, HCMC 2021

Gold Medal, 9th Design, Manufacturing & Application Competition, Ho Chi Minh Communist Youth Union 2021

Champion, Digital Water Hackathon, Singapore University of Social Sciences & Ripple2wave 2021

Champion, AI APP Hackathon, Anscenter, DeepHub & Magik Coworking Space 2020

INDUSTRY EXPERIENCE

AI Engineer Intern

Fujairah Research Center

Fujairah, UAE

Jun–Jul 2024

- Built an internal RAG assistant, integrating retrieval, prompt orchestration, backend logic, and response-generation workflows for organizational knowledge access.

AI Engineer

VinBigData JSC, VinGroup

Hanoi, Vietnam

Jul 2022–May 2023

- Developed a multilingual fine-grained NER system for SemEval-2023 MultiCoNER II using mDeBERTa and self-annotation, improving English and Bangla baselines by 2.38% and 11.33%.

SKILLS

Programming/ML: Python, PyTorch, Transformers, NumPy, Pandas.

Research: Mechanistic interpretability, sparse autoencoders, causal interventions, representation geometry, model steering.